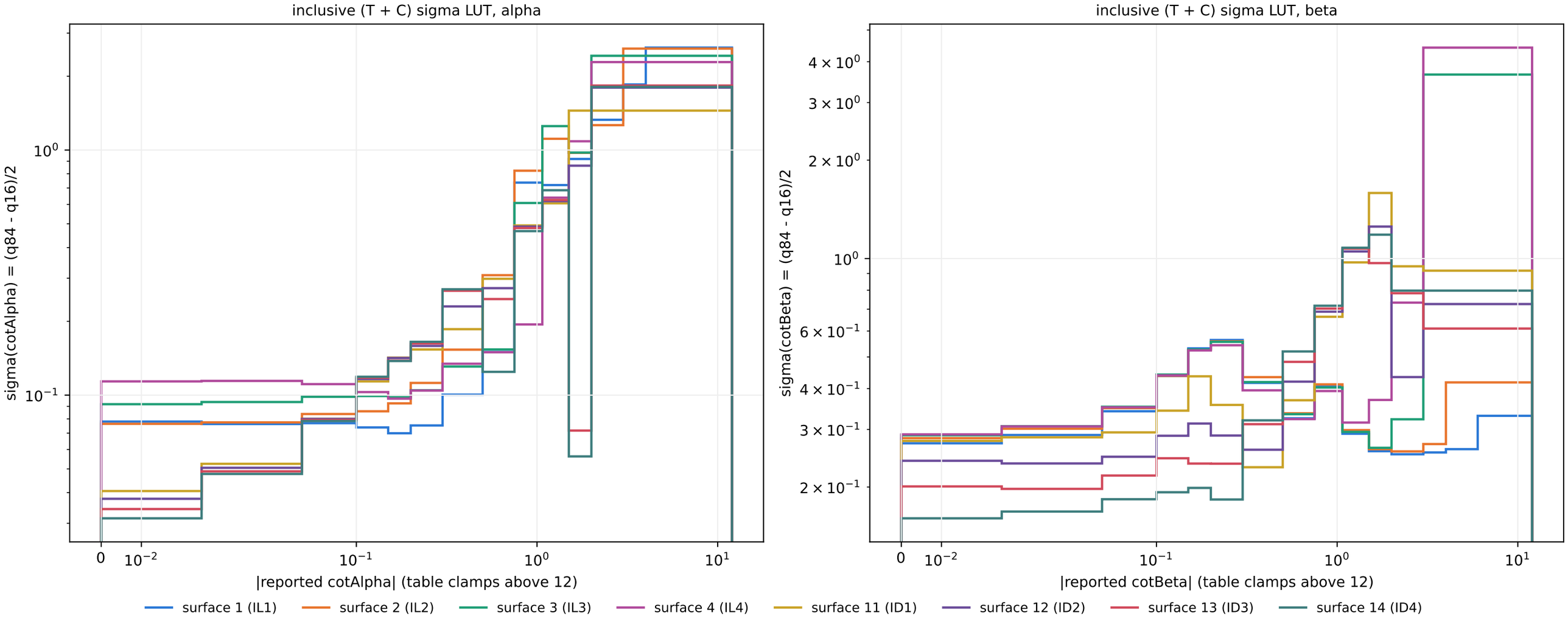

inclusive (T + C) sigma LUT, alpha
sigma(cotAlpha) = (q84 - q16)/2
10^{0}
10^{-1}
0
10^{-2}
10^{-1}
10^{0}
10^{1}
|reported cotAlpha| (table clamps above 12)
inclusive (T + C) sigma LUT, beta
sigma(cotBeta) = (q84 - q16)/2
4×10^{0}
3×10^{0}
2×10^{0}
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
0
10^{-2}
10^{-1}
10^{0}
10^{1}
|reported cotBeta| (table clamps above 12)
surface 1 (IL1)
surface 2 (IL2)
surface 3 (IL3)
surface 4 (IL4)
surface 11 (ID1)
surface 12 (ID2)
surface 13 (ID3)
surface 14 (ID4)

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (1)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
alpha	1	0	0.025	71412	72127	0.2008	0.07799	0.07394	3.031	0.07195	0.109
alpha	1	0.025	0.05	70811	71993	0.2018	0.07623	0.07079	1.111	0.06941	0.1077
alpha	1	0.05	0.1	138912	139019	0.1993	0.07685	0.07219	8.161	0.07066	0.1071
alpha	1	0.1	0.15	126011	126561	0.1999	0.07382	0.06817	1.636	0.067	0.1024
alpha	1	0.15	0.2	127359	126591	0.1982	0.0699	0.06274	1830	0.06382	0.09631
alpha	1	0.2	0.3	200658	199675	0.1984	0.07521	0.06485	3.156	0.06955	0.1043
alpha	1	0.3	0.5	127680	127263	0.1987	0.1004	0.08783	0.6757	0.09128	0.1455
alpha	1	0.5	0.75	23204	23012	0.1979	0.1501	0.07708	0.6432	0.1171	0.3122
alpha	1	0.75	1.07	5860	5726	0.1955	0.7363	0.06558	5.166	0.6825	0.9107
alpha	1	1.07	1.5	3271	3313	0.2012	0.7195	0.06491	2.557	0.6369	1.353
alpha	1	1.5	2	1684	1726	0.2031	0.9193	0.08876	2.902	0.8788	1.761
alpha	1	2	3	1392	1410	0.2012	1.329	0.1535	2.186	1.277	2.557
alpha	1	3	4	576	567	0.1967	1.851	0.2849	2.89	1.785	3.542
alpha	1	4	12	994	760	0.1598	2.617	0.9783	6.064	1.425	7.272
alpha	2	0	0.025	71752	71466	0.286	0.07637	0.07321	0.2181	0.06864	0.1023
alpha	2	0.025	0.05	66578	66506	0.2865	0.07742	0.07364	0.2168	0.06874	0.1042
alpha	2	0.05	0.1	121505	121625	0.287	0.08375	0.08026	0.188	0.07519	0.1103
alpha	2	0.1	0.15	104524	103783	0.2853	0.08593	0.08069	0.2494	0.07601	0.1157
alpha	2	0.15	0.2	85379	84860	0.2855	0.09254	0.08496	0.2229	0.08027	0.1266
alpha	2	0.2	0.3	109548	109714	0.2871	0.1122	0.1093	0.224	0.09872	0.1525
alpha	2	0.3	0.5	65081	66456	0.2911	0.1533	0.1142	0.2944	0.1226	0.2235
alpha	2	0.5	0.75	12482	12337	0.2844	0.3086	0.104	0.468	0.2085	0.4422
alpha	2	0.75	1.07	3458	3499	0.2892	0.8235	0.08338	0.86	0.77	0.9057
alpha	2	1.07	1.5	1445	1404	0.2809	1.111	0.07616	1.232	0.7665	1.319
alpha	2	1.5	2	805	870	0.3029	0.9738	1.759	1.461	0.9365	1.701
alpha	2	2	3	607	671	0.3077	1.264	0.1519	1.961	1.213	2.458
alpha	2	3	12	585	507	0.2584	2.593	0.4044	4.047	2.067	5.298
alpha	3	0	0.025	37243	37017	0.3641	0.09183	0.08351	0.2667	0.0782	0.1299
alpha	3	0.025	0.05	35577	35225	0.3632	0.09375	0.08483	0.6247	0.07884	0.1334
alpha	3	0.05	0.1	67361	67003	0.3642	0.09847	0.09148	0.2993	0.08327	0.1342
alpha	3	0.1	0.15	63426	63780	0.3668	0.09919	0.09189	8.066	0.08421	0.1345
alpha	3	0.15	0.2	58042	57543	0.3635	0.09808	0.08798	0.1945	0.08127	0.1346
alpha	3	0.2	0.3	88797	88489	0.3647	0.1047	0.09446	0.3462	0.0862	0.1464
alpha	3	0.3	0.5	80683	81756	0.3686	0.1308	0.1058	1.121	0.09885	0.2053

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (2)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
alpha	3	0.5	0.75	21304	21154	0.3638	0.1536	0.1063	0.4208	0.1058	0.3455
alpha	3	0.75	1.07	4830	5124	0.3793	0.6081	0.1067	1.525	0.1953	0.8021
alpha	3	1.07	1.5	1192	1347	0.3943	1.252	0.09249	1.13	0.8891	1.345
alpha	3	1.5	2	594	588	0.3631	0.9763	1.749	5.638	0.9359	1.737
alpha	3	2	12	569	574	0.3675	2.424	0.3191	4.369	1.484	4.078
alpha	4	0	0.025	21917	21610	0.4282	0.1138	0.1038	0.2046	0.09002	0.1612
alpha	4	0.025	0.05	21033	21031	0.4316	0.1143	0.1024	0.1891	0.08839	0.1602
alpha	4	0.05	0.1	41891	41868	0.4315	0.1109	0.1012	0.2201	0.08636	0.1568
alpha	4	0.1	0.15	43671	43659	0.4316	0.1029	0.0895	0.2052	0.08114	0.1445
alpha	4	0.15	0.2	42943	43026	0.4321	0.09674	0.08231	0.2292	0.07609	0.1358
alpha	4	0.2	0.3	71269	71114	0.4311	0.1044	0.09385	0.4477	0.08311	0.1416
alpha	4	0.3	0.5	76638	77167	0.4333	0.1343	0.1105	0.278	0.09693	0.2015
alpha	4	0.5	0.75	25889	26198	0.4346	0.1496	0.1072	0.4346	0.0972	0.3495
alpha	4	0.75	1.07	9996	9674	0.4236	0.1943	0.103	0.5695	0.1033	0.6434
alpha	4	1.07	1.5	3313	3032	0.4101	0.6393	0.09393	0.9042	0.1242	1.218
alpha	4	1.5	2	687	848	0.4839	1.086	0.09025	1.568	0.8916	1.783
alpha	4	2	12	594	560	0.4173	2.285	0.2007	3.141	1.219	3.911
alpha	11	0	0.025	180416	180004	0.2496	0.0406	0.037	0.5416	0.03904	0.04764
alpha	11	0.025	0.05	162954	163463	0.2507	0.05245	0.05204	0.4778	0.04989	0.06092
alpha	11	0.05	0.1	274801	275342	0.2504	0.08	0.08291	0.3755	0.07732	0.08993
alpha	11	0.1	0.15	176682	176391	0.2498	0.1138	0.1168	0.462	0.1082	0.1323
alpha	11	0.15	0.2	84754	84493	0.2495	0.1422	0.1495	0.6392	0.1372	0.1736
alpha	11	0.2	0.3	53159	53025	0.2496	0.1535	0.1697	0.8015	0.1348	0.2141
alpha	11	0.3	0.5	20197	20248	0.2505	0.1859	0.1219	0.3575	0.1686	0.3172
alpha	11	0.5	0.75	7512	7496	0.2497	0.2983	0.5239	0.481	0.2861	0.4912
alpha	11	0.75	1.07	3470	3469	0.25	0.4923	0.9678	2.294	0.4894	0.6047
alpha	11	1.07	1.5	2199	2186	0.249	0.6064	1.117	2.902	0.6034	0.1476
alpha	11	1.5	12	1164	1197	0.2554	1.448	0.1503	2.85	0.906	3.382
alpha	12	0	0.025	184774	184526	0.2729	0.03776	0.03368	0.2283	0.03579	0.04314
alpha	12	0.025	0.05	173405	173796	0.2736	0.05052	0.05114	0.1403	0.04818	0.0571
alpha	12	0.05	0.1	307557	307896	0.2734	0.07998	0.07472	0.1561	0.07745	0.08799
alpha	12	0.1	0.15	185013	184923	0.2731	0.1163	0.1037	0.2611	0.1105	0.1315
alpha	12	0.15	0.2	80860	80713	0.2728	0.1418	0.1601	0.1769	0.1327	0.1721
alpha	12	0.2	0.3	45399	45096	0.2718	0.1588	0.2421	0.2299	0.1358	0.2218

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (3)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
alpha	12	0.3	0.5	14572	14770	0.2758	0.2301	0.5002	0.3491	0.1747	0.3579
alpha	12	0.5	0.75	8694	8732	0.274	0.2732	0.5293	0.549	0.2707	0.3077
alpha	12	0.75	1.07	3785	3784	0.2731	0.487	0.9694	0.8299	0.486	0.5277
alpha	12	1.07	1.5	1968	1906	0.2669	0.6186	1.156	1.119	0.6038	1.075
alpha	12	1.5	2	891	775	0.2464	0.8632	1.661	1.559	0.8576	1.166
alpha	12	2	12	814	793	0.268	1.797	0.2155	3.073	0.7114	4.264
alpha	13	0	0.025	181519	181733	0.2979	0.03429	0.03112	0.3844	0.03317	0.03904
alpha	13	0.025	0.05	183866	183789	0.2976	0.04885	0.05042	0.3533	0.04684	0.0538
alpha	13	0.05	0.1	336524	336914	0.2979	0.07919	0.0759	0.2773	0.0768	0.08561
alpha	13	0.1	0.15	187752	187338	0.2972	0.1178	0.106	0.3645	0.1128	0.1309
alpha	13	0.15	0.2	73559	73755	0.2982	0.1383	0.1684	0.1796	0.1269	0.1694
alpha	13	0.2	0.3	36996	36772	0.2964	0.1625	0.1948	0.7782	0.1353	0.2283
alpha	13	0.3	0.5	10956	10900	0.2966	0.2665	0.5004	0.3025	0.211	0.3946
alpha	13	0.5	0.75	10067	10157	0.2995	0.2466	0.5348	0.5315	0.252	0.07773
alpha	13	0.75	1.07	3865	3770	0.2925	0.4807	0.965	0.8708	0.4831	0.3772
alpha	13	1.07	1.5	1434	1378	0.2894	0.6257	1.115	1.061	0.5852	1.216
alpha	13	1.5	2	2138	2144	0.2983	0.07167	1.686	1.664	0.2531	0.06885
alpha	13	2	12	883	867	0.2939	1.832	0.235	2.89	1.383	4.037
alpha	14	0	0.025	179605	179050	0.3215	0.03143	0.02691	0.2333	0.03055	0.03395
alpha	14	0.025	0.05	189078	188874	0.322	0.0477	0.05013	0.2264	0.04651	0.05112
alpha	14	0.05	0.1	356609	357172	0.3226	0.07841	0.07591	0.1262	0.0765	0.08351
alpha	14	0.1	0.15	187133	187918	0.3231	0.1188	0.1074	0.1427	0.1136	0.1299
alpha	14	0.15	0.2	66628	66628	0.3222	0.1378	0.1681	0.1975	0.1251	0.1683
alpha	14	0.2	0.3	29576	29276	0.32	0.165	0.1952	0.2169	0.1345	0.2313
alpha	14	0.3	0.5	8854	8894	0.3232	0.2703	0.5001	0.3317	0.2264	0.4085
alpha	14	0.5	0.75	12394	12178	0.3184	0.1244	0.54	0.9273	0.2158	0.0415
alpha	14	0.75	1.07	3937	3974	0.3243	0.467	0.9615	0.9057	0.4725	0.187
alpha	14	1.07	1.5	931	908	0.3168	0.6855	0.07211	1.056	0.5914	1.299
alpha	14	1.5	2	4173	4104	0.3186	0.05623	1.694	1.652	0.0541	0.06147
alpha	14	2	12	1039	950	0.303	1.81	0.2297	2.971	1.464	3.62
beta	1	0	0.025	17173	16991	0.1975	0.2722	0.2351	0.7542	0.2361	0.4994
beta	1	0.025	0.05	15927	16129	0.2012	0.2885	0.2556	0.6003	0.2529	0.5099
beta	1	0.05	0.1	25878	25739	0.1984	0.3411	0.3182	0.7709	0.3117	0.5296
beta	1	0.1	0.15	18009	17817	0.1975	0.4372	0.4128	0.6655	0.417	0.5865

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (4)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
beta	1	0.15	0.2	12824	12925	0.2005	0.531	0.4962	0.7933	0.5154	0.6507
beta	1	0.2	0.3	18224	18283	0.1997	0.5634	0.5309	1.223	0.5345	0.7133
beta	1	0.3	0.5	38908	38853	0.199	0.4165	0.3767	0.9103	0.36	0.6902
beta	1	0.5	0.75	70116	70119	0.1992	0.3245	0.2864	0.6933	0.2751	0.7042
beta	1	0.75	1.07	75413	75971	0.2004	0.4058	0.3553	0.9186	0.3444	0.7655
beta	1	1.07	1.5	86212	86768	0.2002	0.2912	0.2365	1.179	0.2481	0.6575
beta	1	1.5	2	84939	85005	0.1993	0.2574	0.2249	0.8932	0.2229	0.5069
beta	1	2	3	137084	138955	0.2014	0.2519	0.2274	0.7664	0.2185	0.4796
beta	1	3	4	103869	105154	0.2012	0.2551	0.2313	6.101	0.2222	0.5035
beta	1	4	6	135228	129338	0.1922	0.2611	0.2344	1.046	0.2213	0.5777
beta	1	6	12	59210	57416	0.1943	0.3303	0.277	1235	0.2685	1.055
beta	2	0	0.025	17211	17289	0.2877	0.2823	0.2431	0.4178	0.2329	0.4702
beta	2	0.025	0.05	16199	16407	0.2894	0.3016	0.2647	0.4267	0.255	0.4824
beta	2	0.05	0.1	26778	26673	0.286	0.3486	0.3252	0.4578	0.3124	0.4888
beta	2	0.1	0.15	18545	18543	0.2868	0.4377	0.4179	0.4791	0.4099	0.5494
beta	2	0.15	0.2	13625	13310	0.282	0.5263	0.4974	0.5538	0.5066	0.6203
beta	2	0.2	0.3	18864	18615	0.2841	0.5582	0.532	0.598	0.5261	0.6518
beta	2	0.3	0.5	39002	39175	0.2877	0.4338	0.4057	0.6217	0.3686	0.6006
beta	2	0.5	0.75	72166	72274	0.2871	0.3362	0.3011	0.4718	0.2787	0.5471
beta	2	0.75	1.07	77654	77650	0.2868	0.4121	0.3754	0.5561	0.3522	0.5846
beta	2	1.07	1.5	88761	89053	0.2874	0.2986	0.2509	0.5821	0.2513	0.475
beta	2	1.5	2	83973	84191	0.2873	0.2611	0.2338	0.481	0.2219	0.3905
beta	2	2	3	118707	118001	0.2856	0.257	0.2391	0.4516	0.2178	0.3873
beta	2	3	4	44648	43529	0.2816	0.2707	0.2432	0.6528	0.2268	0.4456
beta	2	4	12	7394	8681	0.3207	0.418	0.278	1.824	0.279	1.138
beta	3	0	0.025	17420	17352	0.3646	0.288	0.2567	0.3573	0.2304	0.4336
beta	3	0.025	0.05	16214	16300	0.3667	0.3069	0.2787	0.4716	0.254	0.4389
beta	3	0.05	0.1	26531	26486	0.3651	0.352	0.3313	0.4078	0.3061	0.4629
beta	3	0.1	0.15	18382	18656	0.3689	0.4415	0.4265	0.5034	0.4142	0.5161
beta	3	0.15	0.2	12906	12848	0.3645	0.5268	0.5062	0.6204	0.5136	0.57
beta	3	0.2	0.3	18492	18716	0.3683	0.5561	0.54	0.547	0.5311	0.6083
beta	3	0.3	0.5	40062	39658	0.3631	0.4189	0.403	0.466	0.3549	0.5319
beta	3	0.5	0.75	70574	70109	0.364	0.3341	0.3079	0.4442	0.2736	0.4726
beta	3	0.75	1.07	74405	74501	0.3658	0.4032	0.3798	1.876	0.344	0.5066

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (5)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
beta	3	1.07	1.5	79189	80222	0.3685	0.2949	0.2521	0.4416	0.2436	0.4114
beta	3	1.5	2	58167	57238	0.3618	0.2636	0.238	0.4756	0.2189	0.3603
beta	3	2	3	25957	25889	0.3649	0.3223	0.2954	0.5361	0.2593	0.4606
beta	3	3	12	1267	1520	0.4087	3.655	0.3973	3.608	3.509	3.843
beta	4	0	0.025	17374	17014	0.4265	0.2898	0.2682	0.3304	0.2324	0.3914
beta	4	0.025	0.05	16089	16149	0.4326	0.3064	0.2881	0.3458	0.2495	0.407
beta	4	0.05	0.1	26022	26200	0.4333	0.3506	0.3347	0.423	0.3048	0.4317
beta	4	0.1	0.15	17993	18234	0.4349	0.439	0.4277	1.41	0.4116	0.4915
beta	4	0.15	0.2	12648	12736	0.4334	0.5236	0.5142	0.4978	0.5098	0.5496
beta	4	0.2	0.3	18405	18087	0.4274	0.5429	0.5329	0.5229	0.525	0.573
beta	4	0.3	0.5	38711	38588	0.4309	0.3951	0.3869	0.4482	0.3425	0.4695
beta	4	0.5	0.75	67321	66752	0.4296	0.3227	0.3032	0.4171	0.2644	0.4151
beta	4	0.75	1.07	66148	66071	0.4314	0.3933	0.3704	0.4873	0.3361	0.4639
beta	4	1.07	1.5	56323	56971	0.4345	0.3149	0.2581	0.4666	0.2542	0.4127
beta	4	1.5	2	19766	19669	0.4304	0.3693	0.3308	0.587	0.2869	0.4917
beta	4	2	3	2434	2685	0.4559	0.7329	0.5789	1.164	0.4575	1.026
beta	4	3	12	569	613	0.45	4.418	0.2012	4.256	4.158	4.84
beta	11	0	0.025	63783	63292	0.2487	0.2768	0.2292	0.4469	0.2724	0.2864
beta	11	0.025	0.05	62664	62762	0.2504	0.2839	0.2356	0.3715	0.2818	0.2886
beta	11	0.05	0.1	124845	124728	0.2499	0.2939	0.2465	0.3569	0.2895	0.3028
beta	11	0.1	0.15	93663	93775	0.2503	0.3425	0.09266	0.4515	0.335	0.3527
beta	11	0.15	0.2	63380	63457	0.2503	0.4362	0.4266	0.4782	0.4367	0.4296
beta	11	0.2	0.3	96507	96787	0.2506	0.3566	0.1018	0.4772	0.3562	0.3549
beta	11	0.3	0.5	162212	162662	0.2506	0.2299	0.1898	0.4089	0.2289	0.235
beta	11	0.5	0.75	175448	175133	0.2498	0.3685	0.3637	0.4503	0.3652	0.3749
beta	11	0.75	1.07	104958	104659	0.2496	0.6635	0.6061	0.6607	0.6644	0.66
beta	11	1.07	1.5	16284	16229	0.2495	0.9732	0.9437	0.8852	0.9696	0.9857
beta	11	1.5	2	1897	1947	0.255	1.587	1.299	1.31	1.195	1.697
beta	11	2	3	756	739	0.2459	0.9466	0.1873	1.501	0.4218	2.545
beta	11	3	12	536	557	0.2574	0.918	0.2036	3.348	0.2618	5.765
beta	12	0	0.025	77441	77544	0.2734	0.2406	0.2139	0.4175	0.2408	0.2421
beta	12	0.025	0.05	76829	77506	0.2749	0.2361	0.2032	0.274	0.2336	0.2425
beta	12	0.05	0.1	157133	156478	0.2723	0.2475	0.2619	0.41	0.245	0.2547
beta	12	0.1	0.15	114195	114738	0.2741	0.287	0.06698	0.327	0.2864	0.2898

sigma LUT bins (training events): sigma = weighted (q84-q16)/2 of the pooled T+C residual [cot]; core/full RMS [cot]; per-route widths; wC = class-C weight share (6)

angle	surface	lo	hi	nT	nC	wC	sigma	core_rms	full_rms	sigma_T_only	sigma_C_only
beta	12	0.15	0.2	69132	69155	0.2732	0.3128	0.3904	0.391	0.3096	0.3203
beta	12	0.2	0.3	98138	97960	0.2728	0.2874	0.106	0.3983	0.2857	0.2896
beta	12	0.3	0.5	139184	138969	0.2729	0.2601	0.2112	0.3509	0.2567	0.27
beta	12	0.5	0.75	151316	151638	0.2736	0.4201	0.3725	0.4153	0.4167	0.4302
beta	12	0.75	1.07	104360	103560	0.2716	0.6878	0.6265	0.6533	0.6873	0.6897
beta	12	1.07	1.5	16891	16980	0.2742	1.051	1.002	0.9963	1.048	1.057
beta	12	1.5	2	1276	1372	0.2878	1.252	0.2075	1.244	0.7871	1.64
beta	12	2	3	652	599	0.2567	0.434	0.1885	1.394	0.2055	2.528
beta	12	3	12	568	521	0.2564	0.7258	0.2135	4.64	0.2559	5.69
beta	13	0	0.025	88697	88672	0.2976	0.2009	0.1646	0.259	0.2	0.2046
beta	13	0.025	0.05	92912	92868	0.2976	0.1974	0.211	0.2056	0.1971	0.2017
beta	13	0.05	0.1	184684	184555	0.2975	0.2169	0.2178	0.278	0.2161	0.2182
beta	13	0.1	0.15	133185	133728	0.2985	0.245	0.0542	0.4744	0.2443	0.2486
beta	13	0.15	0.2	73575	73665	0.2979	0.236	0.3564	0.3508	0.2348	0.2393
beta	13	0.2	0.3	93773	93222	0.2964	0.2358	0.1089	0.4369	0.2345	0.2366
beta	13	0.3	0.5	115113	115429	0.2982	0.3113	0.3267	0.378	0.3112	0.3121
beta	13	0.5	0.75	126838	127035	0.298	0.483	0.4453	0.4976	0.4814	0.4872
beta	13	0.75	1.07	98039	97975	0.2975	0.703	0.6375	0.7564	0.7013	0.7085
beta	13	1.07	1.5	19809	19533	0.2947	1.072	1.034	1.001	1.071	1.076
beta	13	1.5	2	1033	991	0.2891	0.9685	0.2377	1.059	0.7232	1.594
beta	13	2	3	510	530	0.3058	0.7835	0.1752	1.663	0.2084	2.559
beta	13	3	12	565	503	0.2739	0.6107	0.2264	3.161	0.2452	5.297
beta	14	0	0.025	103978	103956	0.3222	0.1604	0.1288	1.818	0.1602	0.1625
beta	14	0.025	0.05	104775	104766	0.3222	0.1683	0.03405	0.1712	0.1685	0.1671
beta	14	0.05	0.1	197200	197830	0.3229	0.1835	0.05662	0.1822	0.1816	0.1852
beta	14	0.1	0.15	149821	149499	0.3217	0.1928	0.2254	0.2213	0.1886	0.1985
beta	14	0.15	0.2	81310	81785	0.3235	0.1987	0.09981	0.2921	0.1987	0.1991
beta	14	0.2	0.3	86057	85861	0.3217	0.1832	0.112	0.3195	0.1766	0.1887
beta	14	0.3	0.5	100750	100320	0.3213	0.3198	0.2722	1.347	0.3187	0.3226
beta	14	0.5	0.75	105723	105842	0.3225	0.5198	0.4739	1.372	0.5182	0.5229
beta	14	0.75	1.07	85452	85875	0.3233	0.7172	0.7069	0.708	0.7145	0.7236
beta	14	1.07	1.5	22101	21506	0.3163	1.079	1.031	1.027	1.078	1.081
beta	14	1.5	2	720	731	0.3255	1.183	0.2051	1.206	0.3052	1.74
beta	14	2	12	1103	1000	0.3012	0.7978	0.18	3.631	0.2161	4.116

Pulls (emitted - true cot)/sigma, python reference on the TEST events (class-T clusters through both routes) (1)

angle	surface	route	selection	n	q_width	median	frac_abs_gt3
alpha	1	C	all	358769	1.417	-0.03108	0.07138
alpha	1	C	|cot|<=12, not saturated	358550	1.416	-0.03111	0.07082
alpha	1	T	all	358769	0.9153	-0.01627	0.02329
alpha	1	T	|cot|<=12, not saturated	358675	0.9149	-0.01625	0.02314
alpha	2	C	all	256788	1.364	-0.0318	0.05016
alpha	2	C	|cot|<=12, not saturated	256734	1.363	-0.03172	0.04996
alpha	2	T	all	256788	0.879	-0.01361	0.00863
alpha	2	T	|cot|<=12, not saturated	256739	0.8788	-0.01361	0.008569
alpha	3	C	all	184420	1.414	-0.03716	0.0831
alpha	3	C	|cot|<=12, not saturated	184369	1.413	-0.03713	0.08286
alpha	3	T	all	184420	0.816	-0.02461	0.0132
alpha	3	T	|cot|<=12, not saturated	184395	0.8159	-0.02455	0.01313
alpha	4	C	all	143607	1.452	-0.03824	0.1114
alpha	4	C	|cot|<=12, not saturated	143567	1.452	-0.03794	0.1111
alpha	4	T	all	143607	0.758	-0.02846	0.01353
alpha	4	T	|cot|<=12, not saturated	143586	0.7578	-0.02838	0.01341
alpha	11	C	all	384539	1.225	-0.1396	0.0331
alpha	11	C	|cot|<=12, not saturated	384500	1.225	-0.1395	0.03301
alpha	11	T	all	384539	1.002	-0.09129	0.007196
alpha	11	T	|cot|<=12, not saturated	384510	1.002	-0.09129	0.007139
alpha	12	C	all	401153	1.2	-0.1928	0.03045
alpha	12	C	|cot|<=12, not saturated	401126	1.2	-0.1928	0.03038
alpha	12	T	all	401153	1.019	-0.1308	0.006436
alpha	12	T	|cot|<=12, not saturated	401143	1.019	-0.1308	0.006437
alpha	13	C	all	409699	1.188	-0.2892	0.03054
alpha	13	C	|cot|<=12, not saturated	409671	1.188	-0.2892	0.03048
alpha	13	T	all	409699	1.034	-0.2175	0.008228
alpha	13	T	|cot|<=12, not saturated	409680	1.034	-0.2173	0.008189
alpha	14	C	all	415256	1.168	-0.3604	0.04067
alpha	14	C	|cot|<=12, not saturated	415209	1.167	-0.3604	0.04056
alpha	14	T	all	415256	1.043	-0.291	0.01974
alpha	14	T	|cot|<=12, not saturated	415199	1.043	-0.2907	0.01963
beta	1	C	all	358769	2.021	-0.0197	0.2017
beta	1	C	|cot|<=12, not saturated	356843	2.001	-0.01904	0.1974

Pulls (emitted - true cot)/sigma, python reference on the TEST events (class-T clusters through both routes) (2)

angle	surface	route	selection	n	q_width	median	frac_abs_gt3
beta	1	T	all	358769	0.868	-0.01538	0.01631
beta	1	T	|cot|<=12, not saturated	358311	0.8665	-0.01516	0.0154
beta	2	C	all	256788	1.503	-0.03255	0.09607
beta	2	C	|cot|<=12, not saturated	256603	1.501	-0.03222	0.09542
beta	2	T	all	256788	0.8582	-0.02397	0.01774
beta	2	T	|cot|<=12, not saturated	256660	0.8574	-0.0238	0.01732
beta	3	C	all	184420	1.333	-0.03857	0.05636
beta	3	C	|cot|<=12, not saturated	184336	1.332	-0.03817	0.05594
beta	3	T	all	184420	0.8486	-0.02145	0.01574
beta	3	T	|cot|<=12, not saturated	184349	0.8482	-0.02138	0.01542
beta	4	C	all	143607	1.228	-0.03131	0.03909
beta	4	C	|cot|<=12, not saturated	143544	1.227	-0.03127	0.03867
beta	4	T	all	143607	0.8486	-0.017	0.01341
beta	4	T	|cot|<=12, not saturated	143562	0.8482	-0.01699	0.01314
beta	11	C	all	384539	1.041	0.01563	0.01227
beta	11	C	|cot|<=12, not saturated	384250	1.041	0.01476	0.01153
beta	11	T	all	384539	1.017	0.01013	0.007583
beta	11	T	|cot|<=12, not saturated	384349	1.017	0.009118	0.007132
beta	12	C	all	401153	1.08	0.04956	0.007628
beta	12	C	|cot|<=12, not saturated	400870	1.079	0.04833	0.006927
beta	12	T	all	401153	1.059	0.01812	0.002343
beta	12	T	|cot|<=12, not saturated	400899	1.058	0.01754	0.001739
beta	13	C	all	409699	1.088	0.1382	0.006666
beta	13	C	|cot|<=12, not saturated	409312	1.087	0.1357	0.005727
beta	13	T	all	409699	1.071	0.1189	0.001362
beta	13	T	|cot|<=12, not saturated	409361	1.07	0.1171	0.0005765
beta	14	C	all	415256	1.109	0.2272	0.009991
beta	14	C	|cot|<=12, not saturated	414847	1.107	0.2252	0.009015
beta	14	T	all	415256	1.094	0.218	0.003653
beta	14	T	|cot|<=12, not saturated	414816	1.093	0.2163	0.002673

Pulls of the CMSSW nano (file 3), class T

angle	surface	cls	selection	n	q_width	median	frac_abs_gt3
alpha	1	T (CMSSW nano)	all	719916	0.9144	-0.01858	0.02326
alpha	1	T (CMSSW nano)	|cot|<=12	719562	0.9139	-0.0184	0.023
beta	1	T (CMSSW nano)	all	719916	0.8675	-0.01357	0.01665
beta	1	T (CMSSW nano)	|cot|<=12	718899	0.8656	-0.0134	0.01562
alpha	2	T (CMSSW nano)	all	516001	0.8751	-0.01418	0.008452
alpha	2	T (CMSSW nano)	|cot|<=12	515941	0.8749	-0.01415	0.008389
beta	2	T (CMSSW nano)	all	516001	0.8559	-0.02192	0.01784
beta	2	T (CMSSW nano)	|cot|<=12	515746	0.8553	-0.02152	0.01743
alpha	3	T (CMSSW nano)	all	369094	0.8169	-0.02285	0.01298
alpha	3	T (CMSSW nano)	|cot|<=12	369028	0.8168	-0.02278	0.01287
beta	3	T (CMSSW nano)	all	369094	0.8514	-0.02727	0.01593
beta	3	T (CMSSW nano)	|cot|<=12	368956	0.851	-0.02723	0.01562
alpha	4	T (CMSSW nano)	all	288972	0.7615	-0.02713	0.01376
alpha	4	T (CMSSW nano)	|cot|<=12	288883	0.7612	-0.02697	0.01352
beta	4	T (CMSSW nano)	all	288972	0.8491	-0.02558	0.01364
beta	4	T (CMSSW nano)	|cot|<=12	288885	0.8488	-0.02558	0.01342
alpha	11	T (CMSSW nano)	all	771650	1.002	-0.08825	0.007325
alpha	11	T (CMSSW nano)	|cot|<=12	771588	1.002	-0.08825	0.007258
beta	11	T (CMSSW nano)	all	771650	1.017	0.008731	0.007512
beta	11	T (CMSSW nano)	|cot|<=12	771269	1.017	0.007616	0.007057
alpha	12	T (CMSSW nano)	all	802953	1.015	-0.1334	0.006613
alpha	12	T (CMSSW nano)	|cot|<=12	802888	1.015	-0.1334	0.006568
beta	12	T (CMSSW nano)	all	802953	1.057	0.02761	0.002519
beta	12	T (CMSSW nano)	|cot|<=12	802370	1.056	0.02653	0.001833
alpha	13	T (CMSSW nano)	all	822387	1.034	-0.2174	0.00831
alpha	13	T (CMSSW nano)	|cot|<=12	822330	1.034	-0.2174	0.008266
beta	13	T (CMSSW nano)	all	822387	1.071	0.1171	0.001364
beta	13	T (CMSSW nano)	|cot|<=12	821708	1.07	0.1158	0.000572
alpha	14	T (CMSSW nano)	all	832463	1.042	-0.294	0.01948
alpha	14	T (CMSSW nano)	|cot|<=12	832427	1.042	-0.294	0.01947
beta	14	T (CMSSW nano)	all	832463	1.093	0.2179	0.003586
beta	14	T (CMSSW nano)	|cot|<=12	831653	1.092	0.2163	0.00265